Credit spreads and the cost of debt capital (1)

- **The cost of debt capital and the market price of corporate bonds depend on the credit risk of a particular bond.**
- **The credit risk therefore influences the credit spread.**
- **The *credit spread* is the *premium required* by an investor in a corporate bond to *compensate for the credit risk* of the bond.**
- **However, yield on risky bonds is affected by the yield on government bonds of the same maturity.**

Credit spreads and the cost of debt capital (2)

- **The yield to a government bondholder is the compensation to the investor for forgoing consumption today and saving.**
- **A government bond is considered to be free of credit risk.**
- ***Therefore the yield on a corporate bond will be:***
- **Yield on corporate bond = risk free rate + credit spread**
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Credit spreads and the cost of debt capital (2)

- **Since credit spreads reflect the credit risk of a bond, they will be inversely related to the credit quality of the bond.**
- **Low credit quality bonds will be characterised by large spreads and high credit quality bonds will be characterized by low spreads.**

Credit spreads by type of bond and maturity

Corporate spreads for Industries (in basis points) 1 basis point = 0.01%							
Rating	1 year	2 year	3 year	5 year	7 year	10 year	30 year
AAA	5	10	15	20	25	33	60
AA	15	25	30	35	44	52	71
A	35	44	55	60	65	73	90
BBB	60	75	100	105	112	122	143
BB	140	180	210	205	210	250	300
B	215	220	260	300	315	350	450
CCC	1,125	1,225	1,250	1,200	1,200	1,275	1,400

Activity 1

- **The current return on a 10-year government bond is 4.2%.**
- **FS Co, a business rated AA, has 10-year bonds in issue.**
- **<u>Required:</u>**
- **Using the credit spread *table above***
- **Calculate the expected yield on FS Co.'s bonds.**

Solution to Activity 1

- **The credit spread on a 10-year AA bond is 52.**
- **This means that 0.52% should be added to the return on the government bond.**
- ***Yield on corporate bond = risk free rate + credit spread***
- **Expected yield on FS Co.'s bonds = 4.2% + 0.52% = 4.72%**

Activity 2

- **A 15-year government bond has a current yield of 5%.**
- **Folks Co, a B-rated company, has an equivalent bonds in issue.**
- **<u>Required:</u>**
- **What is the expected yield on Folks Co.'s bonds?**

Solution to activity 2

- The table does not include credit spreads for 15-year bonds.
- Thus available figures have to be adjusted.
- The credit spread on a 10-year B-rated bond is 350 and a 30-year B-rated bond is 450.
- The adjustment can be calculated as follows.
- $350 + \frac{(450 - 350)}{(30 - 10)} \times 5 = 375$ basis points
- Thus 3.75% must be added to the yield on government bonds.
- The expected yield on Folks' bonds is 5% + 3.75% = 8.75%

The cost of debt capital

- The cost of debt capital for a company is determined using the the following elements.
- (a) Its credit rating
- (b) The maturity of the debt
- (c) The risk-free rate at the appropriate maturity
- (d) The corporate tax rate
- Cost of debt capital = (1 – tax rate)(risk-free rate + credit spread)

Activity 3

- **MP Co, a BBB-rated company, has 5-year bonds in issue.**
- **The firm is 25% geared while the return on the market is 10.65%**
- **The current yield on equivalent government bonds is 3.7%.**
- **Equity beta is 1.05 while corporate tax is at 28%.**
- ***Using the information in the credit spread table above, calculate:***
- **(a) MP Co.'s post-tax cost of debt associated with these 5-year bonds**
- **(b) Calculate MP Co.'s overall cost of capital.**

Solution to activity 3

- (a) **Expected yield on 5-year bonds = *3.7% + 1.05% = 4.75%***
- ***Post-tax cost of debt = Cost of debt before tax *(1 – tax rate)***
- **Post-tax cost of debt = *(1 – 0.28) *4.75 = 3.42%***
- ***(b)* Cost of equity = rf + β*(rm –rf)**
- ***= 3.7% + 1.05*(10.65% - 3.7%)***
- ***= 11.00%***
- ***WACC = (11.00%*0.75) + (3.42%*0.25)***
- ***=8.25% + 0.855%***
- ***= 9.105% = <u>9.11%</u>***

Solution to activity 3

- (a) *WACC = (11.00%*0.75) + (3.42%*0.25)*
- *=8.25% + 0.855%*
- *= 9.105% = <u>9.11%</u>*

Capital element	*Cost %*	*Proportion*	*Cost*Prop*
Equity	*11*	*0.75*	*8.250*
Debt	*3.42*	*0.25*	*<u>0.855</u>*
		WACC	*<u>9.105</u>*